

OE-II-405 (A)

Examination –May- 2024

B.Tech. IV Sem: Common to all Branches Foundation of Block Chain Technologies

Time : 3 Hrs

Max. Marks : 60

Min. Marks : 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

	Marks	BL	CO
Q.1 (a) What is the genesis block?	02	1	CO1
i) The last block created in the Blockchain			
ii) Any block created by the founder			
iii) The first block of a Blockchain			
iv) The first transaction in each block			
(b) What can you find in the block of a blockchain?	02	1	CO1
i) Timestamp ii) Transaction data			
iii) Hash point iv) All of the above			
(c) Define Block chain .How it differs from traditional system?	03	2	CO1
OR			
What is the role of a Hash Puzzle in a Block chain system?	03	2	CO1
(d) Define how Block chain is changing the landscape of digitization?	05	2	CO1
OR			
Explain in detail about Public Key Cryptosystem and how it differs from Private key Cryptosystem ?	05	2	CO1
Q.2 (a) What is a node?	02	1	CO2
i) A type of cryptocurrency ii) A Blockchain			
ii) A computer on a Blockchain network iv) An exchange			
(b) Are blockchain fully public?	02	1	CO2
i) Yes ii) No			
iii) It depends iv) None of the above			
(c) Explain in detail Proof of Work with a diagram ?	03	2	CO2

OR
Explain in detail Proof of Stake with a diagram ?

03 2

- (d) Draw architecture of Block Chain. Explain each layer with its components in detail? 05 6 CO2

OR

What is Smart Contract. Define challenges and main features of Smart Contract?

05 3 CO2

Q.3 (a) Bitcoin is created by _____.

02 1 CO3

(b) Which of the following is not a blockchain component?

02 1 CO3

- i) certificate authority ii) Ledger
iii) Wallet iv) Hash

(c) Differentiate between Permission & Permission less Block chains?

03 4 CO3

OR

Explain how blockchain accumulates blocks.

03 2 CO3

(d) Define different types of Block chain networks based on Authority & Access with diagrams?

05 3 CO3

OR

Why is Block chain protocol important to crypto ? What are the main types of Block chain Protocols ?

05 2 CO3

Q.4 (a) A Bitcoin blockchain uses?

02 1 CO4

- i) RSA ii) SHA-128
iii) SHA-256 iv) All of the above

(b) What is cold storage?

02 1 CO4

- i) A place to hang your coat
ii) A private key connected to the Internet
iii) A private key not connected to the Internet
iv) A desktop wallet

(c) What is the role of Bitcoin in the Block chain technology ?

03 2 CO4

OR

What are the two types of accounts that exist in Ethereum?

03 2 CO4

(d) How is Ethereum private blockchain different from Hyperledger fabric?

05 2 CO4

OR

	Explain art of debugging?	03	II	CO4
(d)	Compare and contrast between Black box testing and White box testing.	05	III	CO4
	OR			
	What do you mean by System testing? Explain in detail.	05	II	CO4
Q.5 (a)	The modification of the software to match changes in the ever changing environment, falls under which category of software maintenance? i) Corrective ii) Adaptive iii) Perfective iv) Preventive	02	II	CO5
(b)	Which of the following risk is the failure of a purchased component to perform as expected? i) Product Risk ii) Project Risk iii) Project Risk iv) Programming Risk	02	II	CO5
(c)	Explain CASE tool.	03	II	CO5
	OR			
	Write the steps of Software Configuration Management.	03	II	CO5
(d)	What do you mean by software maintenance? Explain	05	II	CO5
	OR			
	Explain RMMM Plan in detail.	05	II	CO5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating
